Physics of the Earth and Planetary Interiors, 52 (1988) 330–338
Elsevier Science Publishers B.V., Amsterdam – Printed in The Netherlands

Palaeomagnetism of the Puerto Vallarta intrusive complex and the accretion of the Guerrero terrain, Mexico

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(Received June 25, 1987; revision accepted October 19, 1987)

Böhnel, H. and Negendank, J.F.W., 1988. Palaeomagnetism of the Puerto Vallarta intrusive complex and the accretion of the Guerrero terrain, Mexico. Phys. Earth Planet. Inter., 52: 330–338.

The batholith exposed around Puerto Vallarta is part of the Guerrero tectonostratigraphic terrain, presumably accreted to Mexico during the Laramide orogenic phase. Fourteen sites on this batholith were studied palaeomagnetically and geochemically. Nine sites define a paleopole at 76.6° N, 181.8° E, with $\alpha_{95} = 6.6°$, coinciding with the reference pole for stable North America. It is proposed that no important movements of the region relative to North America have occurred since emplacement of the intrusive rocks ~ 80 Ma ago. This would mean that the Guerrero terrain was accreted to the Mexican continental margin before that time. Geochemical data support this interpretation, characterizing the rocks as volcanic-arc granites intruded at an active continental margin.

1. Introduction

Since at least 100 Ma ago, convergence has been a dominant tectonic process between the Pacific plates and North America, including Mexico (de Cserna, 1960; King, 1969; Coney, 1978; Jurdy, 1984; Engebretson et al., 1985). Nevertheless, the Middle America Trench lacks the characteristics of other trenches in arc systems with a long subduction history (Karig and Sharman, 1975; Karig et al., 1978). It cuts old structural trends at high angles (King, 1969; de Cserna, 1971; Kesler, 1973) and is situated much too close to the Cretaceous plutons along the Mexican coast, if these represent the roots of an eroded ancient volcanic arc. Different processes have been proposed to explain this situation: subduction of continental crust ('tectonic erosion') (Scholl et al., 1970; Katz, 1971), or lateral offset associated with oblique subduction of the Farallon Plate under North America (de Cserna, 1971; Malfait and Dinkelman, 1972; Karig, 1974; Karig et al., 1978; Beck and Plumley, 1979; Beck, 1986). Palaeomagnetic results from Honduras (Chortis block) may point to left-lateral displacement along the Middle America Trench (Fig. 1a) (Gose and Swartz, 1977; Carfantan, 1983; Anderson and Schmidt, 1983). Beck and Plumley (1979) and Beck (1986), on the other hand, propose right-lateral and therefore northward displacement of many Mesozoic batholiths. Models for the subduction of the Farallon Plate beneath North America as derived from relative plate motions

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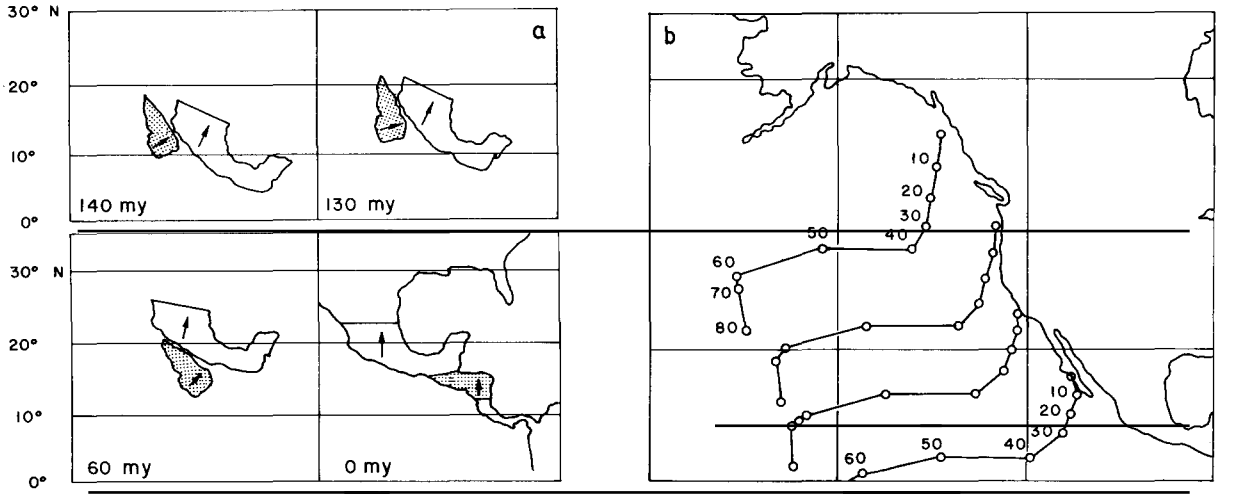


Fig. 1. (a) Relative position of the Chortis block and Mexico as proposed by Gose and Swartz (1977). Intermediate positions are not shown. The longitudinal position of Chortis relative to Mexico is not constrained by the palaeomagnetic data. (b) Trajectory of four points on the Farallon Plate relative to North America during the past 80 Ma as calculated from relative plate motions, treating Antarctica as two plates (Jurdy, 1984).

(Jurdy, 1984) favour right-lateral displacement for most of the last 80 Ma but between about 60 and 40 Ma fast convergence with left-lateral components may have occurred along the Mexican continental margin (Fig. 1b). The situation is still further complicated by the fact that the course of the now subducted spreading ridge once separating the Farallon and Kula plates is not well known, and it is uncertain when, where, and at which angle this ridge collided with North America (Fig. 2). It could well be possible that along the Mexican palaeomargin right-lateral and left-lateral displacements occurred simultaneously, although this depends largely on the actual contour of the margin during that period.

Large parts of western Mexico belongs to the Guerrero terrain (Fig. 3). This suspect terrain is at least Jurassic to mid-Cretaceous in age and composed mainly of submarine volcanic and sedimentary sequences, supposedly accreted to North

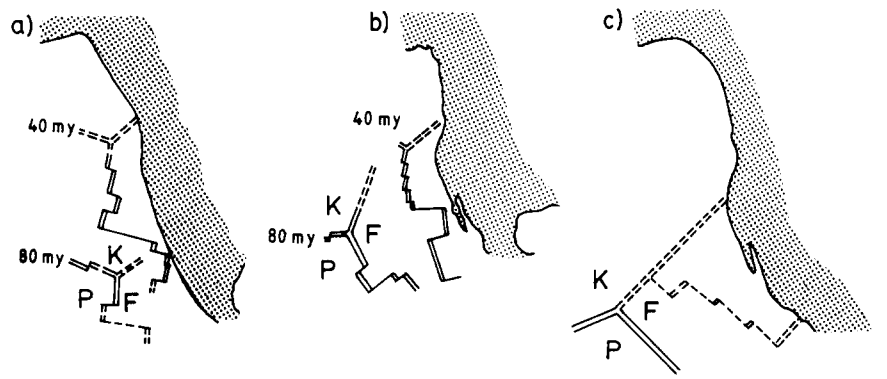


Fig. 2. Late Cretaceous and early Tertiary plate configuration of the eastern Pacific, after (a) Atwater (1970), (b) Coney (1978), and (c) Engebretson et al. (1985). Modified from Beck (1986). P, K, and F are Pacific, Kula, and Farallon plates; spreading ridges (transform faults) are shown by double (single lines); broken lines indicate that the course is inferred.

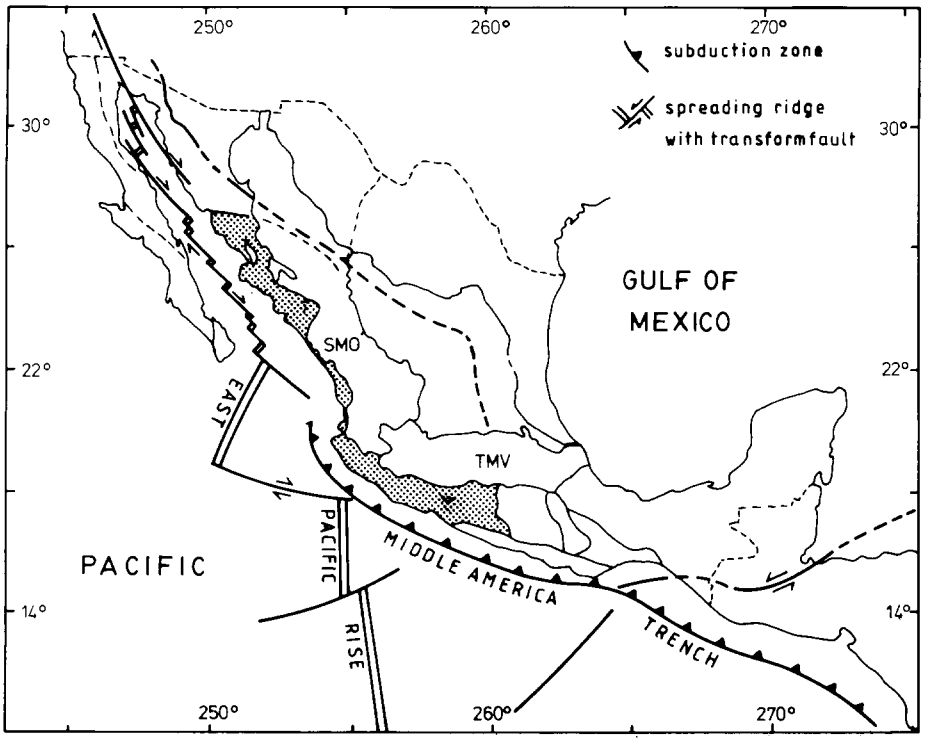


Fig. 3. Tectonic features and the subdivision of Mexico into tectonostratigraphic terrains according to Campa and Coney (1983). Sierra Madre Occidental (SMO) and Trans-Mexican Volcanic Belt (TMV) are overlapping terrains built up from Tertiary and Quaternary volcanics. The Guerrero terrain is shaded.

America by late Mesozoic to early Tertiary time (Campa and Coney, 1983).

2. Geology and geochronology

The Puerto Vallarta intrusive complex is situated in the central part of the Guerrero terrain (Fig. 4). Rock composition ranges from gabbro through diorite and granodiorite to granite. According to the classification of Chappel and White (1974), they are I-type granites of calc-alkaline to high-K calc-alkaline character, containing biotite and hornblende as the dominant ferromagnesian minerals. Magnetite and titanomagnetite are present; ilmenite is subordinate or absent. The diagram Rb versus Y + Nb, after Pearce et al. (1984), reveals that these rocks are similar to granitoid rocks from Chile belonging to 'volcanic-

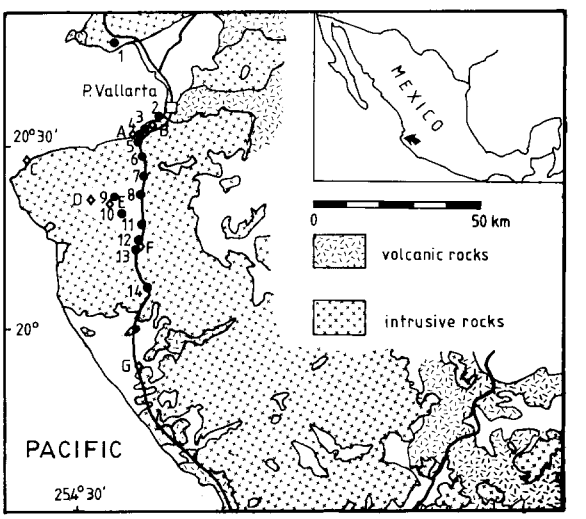


Fig. 4. Geological sketch map of the study area. Sites for palaeomagnetic sampling (●) are marked with numbers from 1 to 14. Radiometric data are known from sites A to G (◇).

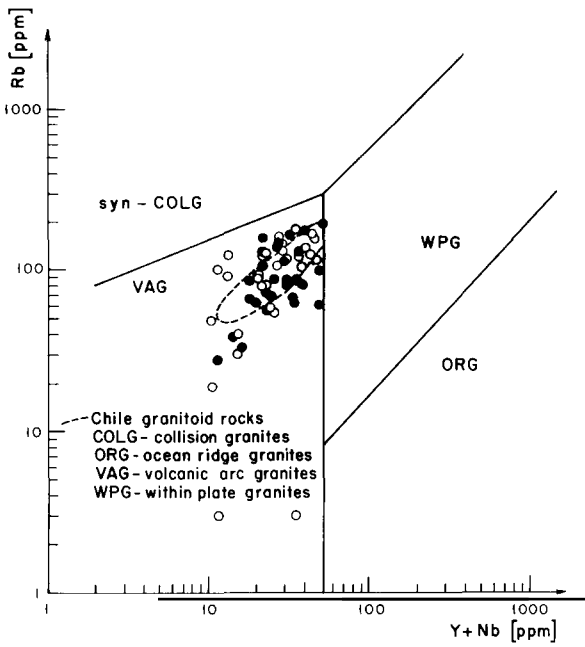


Fig. 5. Diagram Rb versus Y + Nb after Pearce et al. (1984) for intrusive rocks between Puerto Vallarta and Acapulco. Data from the study area are shown as filled circles.

arc granites' intruded at an active continental margin (Fig. 5).

Several samples from the Puerto Vallarta region have been studied geochronometrically by Gastil et al. (1976) and Grajales-Nishimura and Lopez-Infanzon (1983). The sampling sites are marked on Fig. 4 and the results are listed in Table I. Data were obtained from whole-rock, hornblende or biotite separates. They could represent different episodes of the intrusion history between emplacement and cooling. Most apparent ages group quite closely around 80 Ma, which may correspond to the time of acquisition of remanence.

3. Sampling and laboratory procedures

Sampling for palaeomagnetic studies was done mainly in road cuts along the National Highway 200 passing by Manzanillo and Puerto Vallarta (Fig. 4). Apart from the highway, sampling was difficult because of limited access and deep weathering. Fourteen sites were sampled, yielding between five and eight cores, drilled with a portable corer with a diamond bit. The samples were oriented with an inclinometer and magnetic compass (e.g., Collinson, 1983) and later cut into 21 mm long specimens.

Magnetization was determined with a Digico magnetometer (Molyneux, 1971). For demagnetization an alternating field (af) and a Schoenstedt TSD-1 thermal demagnetizer were used. Susceptibility was measured with low-field bridges (e.g., Collinson, 1983). Low-temperature studies of remanence were performed with a modified Digico spinner magnetometer. Curie temperatures were obtained from thermomagnetic curves recorded with a horizontal Curie balance (e.g., Collinson, 1983).

TABLE I

Geochronological results from the Puerto Vallarta intrusive (site code refers to Fig. 4)

Site	Rock composition	Age (Ma)	Dated mineral	Reference
A	Tonalite	80.2 ± 2.2	Biotite	Gastil et al. (1976)
		80.2 ± 2.2	Hornblende	
B	Granodiorite	81.6 ± 2.1	Biotite	Gastil et al. (1976)
		90.9 ± 2.3	Hornblende	
C	Tonalite	80.5 ± 2.1	Biotite	Gastil et al. (1976)
		81.8 ± 2.1	Hornblende	
D	Diorite	78 ± 6	Biotite	Grajales-Nishimura and Lopez-Infanzon (1983)
E	Hypabyssal porphyry	80.6 ± 2.1	Hornblende	Gastil et al. (1976)
F	Tonalite	82.2 ± 0.8	Hornblende	Gastil et al. (1976)
G	Tonalite	93 ± 7	Biotite	Grajales-Nishimura and Lopez-Infanzon (1983)

4. Ore petrology and rock magnetism

In order to obtain information concerning the stability of remanent magnetization, ore petrological studies and rock magnetic experiments were carried out.

Opaque minerals were studied with a conventional reflecting-light microscope and a scanning electron microscope (SEM) equipped with a microanalyser for the semi-quantitative determination of atomic composition. Most sites contain magnetite or titanomagnetite with grain size > 30 μm (Table II). Site 1 is characterized by pyrite with small titanomagnetite inclusions, site 10 by haematite, produced by low-temperature oxidation. The effective grain-size is sometimes reduced due to such effects as maghemitization, hydrothermal alterations, and intergrowth of magnetite with non-magnetic phases.

Magnetic stability indicators Q (Königsberger factor) and MDF (medium destructive field) point to an important contribution from multidomain (MD) particles, which is in good agreement with the observed grain sizes (e.g., Stacey and Banerjee, 1974). Thermomagnetic curves are reversible or nearly so, with Curie temperatures typical for magnetite (Table II). The variation of isothermal remanent magnetization during cooling of the specimen to $-196\,^{\circ}$C and subsequent rewarming to room temperature may also be interpreted as resulting from the dominance of MD magnetite particles, accompanied by much smaller amounts of titanomagnetite and haematite.

From the rock magnetic experiments and from microscopic observations it appears that the magnetization of most specimens is dominated by MD magnetite, which may be susceptible to acquiring viscous remanent magnetization during their history and spurious magnetizations during laboratory treatment. A small contribution of single domain (SD) or pseudo-SD particles may be present as a result of maghemitization, hydrothermal alterations, and mineral intergrowth. This component may still preserve the original remanence acquired during cooling of the intrusive.

5. Paleomagnetic results

Specimens were stored and studied in a reduced magnetic field of about 1–5 μT to prevent acquisi-

TABLE II

Results from petrological and rock magnetic studies

Site no.	Ore petrology				Rock magnetic properties					
	Dominant ore phase	d (μm)	Alteration	Subordinate ores	M_0 (mA m^{-1})	χ (10^{-6})	Q	MDF (kA m^{-1})	T_c/T_c' (°C)	IRM$_{-196}$/IRM$_{20}$
1	Pyrite	<150		TiMa traces	5.50± 2.40	0.40±0.09	0.4	>60		
2	TiMa 3	<500	hydr.		62.4 ± 36.9	2.39±0.55	0.7	3.0± 1.8		10/34
3	n.d.				37.8 ± 22.7	2.44±1.15	0.4	4.2± 0.6	580/580	18/34
4	n.d.				7.35± 2.15	2.96±0.46	0.1	20.6±16.2		
5	Magnetite	c. 30		Il, TiMa, Py	19.1 ± 11.1	11.0 ±0.41	0.5	13.5± 7.0	575/575	14/36
6	n.d.				24.1 ± 13.0	1.68±0.41	0.4	9.6± 19.3	595/600	
7	Magnetite	60–100	magh.	Il, Py	88.9 ± 41.5	7.65±0.95	0.3	3.8± 1.0	570/565	14/28
8	Magnetite	<1000		Il, Il–He, TiMa	562 ±213	32.6 ±5.2	0.5	4.5± 5.0	600/600	10/18
9	n.d.				62.3 ± 30.5	3.57±2.48	0.5	8.2± 2.8		
10	Hematite	20–100	hydr.		27.4 ± 18.0	1.27±0.80	0.6	9.9± 2.0		35/64
11	Magnetite	30–300		Il, Py	290 ±111	23.7 ±2.5	0.3	4.3± 1.4	600/600	14/27
12	Magnetite	30–200		Il, Py	135 ± 59	9.94±3.11	0.4	2.6± 1.0	565/565	10/12
13	Magnetite	30–200		Il, Py	31.2 ± 14.5	4.72±0.35	0.2	3.4± 0.4	575/570	14/17
14	n.d.				44.4 ± 14.0	9.50±0.89	0.1	1.4± 0.6		

TiMa, Titanomagnetite; Il, ilmenite; Py, pyrite; Il–He, ilmenite–haematite; hydr., hydrothermal; magh., maghemitized; M_0, NRM-intensity; χ, magnetic susceptibility; O, Königsberger factor; MDF, medium destructive field; T_c/T_c', Curie temperature from heating/cooling branch of thermomagnetic curve; IRM$_{-196}$/IRM$_{20}$ = percentage of original IRM after cooling to $-196\,^{\circ}$C and after rewarming to room temperature.

tion of viscous remanent magnetization (VRM) and to relax VRMs acquired in former times. Most specimens were af-demagnetized in six or more steps up to ~ 25 mT; in certain cases up to 100 mT. To determine the characteristic remanence, vector plots (As, 1960; Zijderveld, 1967) and stereographic projections of directions and removed components between demagnetization steps (Hoffman and Day, 1978) were evaluated. Many specimens exhibited simple demagnetization curves, as shown in Fig. 6a–c, but sometimes the behaviour was rather unstable (Fig. 6d), especially for samples from sites 12 and 14. The poor directional stability coincides with low values of Q

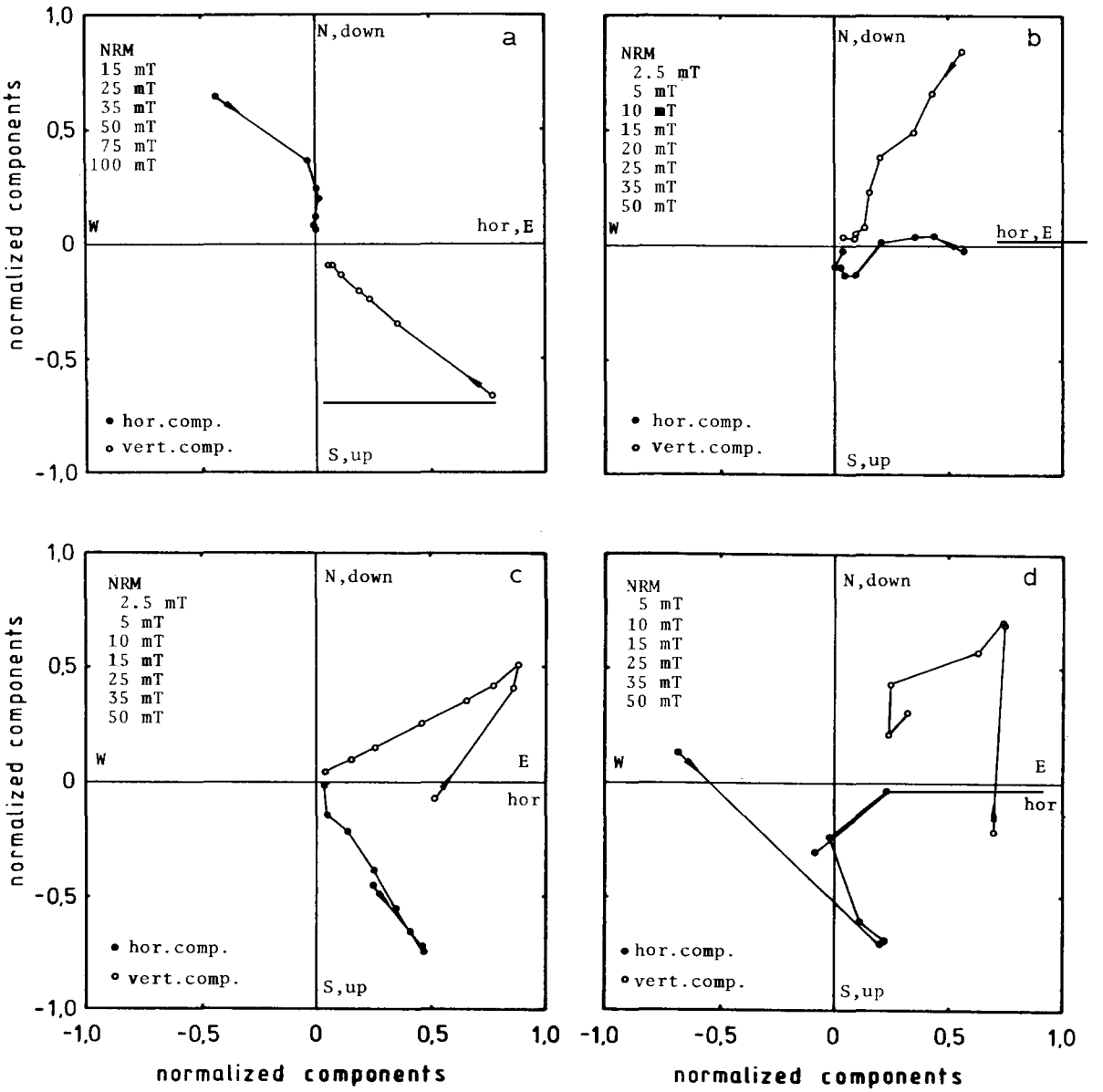


Fig. 6. Alternating field demagnetization curves (As, 1960; Zijderveld, 1967). See text for explanation.

and MDF and may be explained in terms of the predominance of MD particles which are more susceptible to viscous effects during the laboratory procedure.

In this way, for most specimens a characteristic direction could be determined and, excluding sites 12 and 14, site mean directions were calculated (Fisher, 1953). These directions and their correspondent virtual geomagnetic poles (VGP) are listed in Table III, together with statistical parameters n, R, k and α_{95}. k generally is > 20 and α_{95} exceeds 20° only for site 7, which may be due to the small number of specimens, $n = 4$. Three sites have reversed polarity and at least these directions are not affected by an important viscous overprint. The occurrence of mixed polarities in the intrusive complex indicates that remanence was acquired at the end of the Cretaceous long normal interval, or after it, which agrees well with the apparent radiometric ages (Table I).

Figure 7 shows the VGPs and the 80 and 70 Ma reference poles for North America according to Irving and Irving (1982) with their 63% error limits. If 63% error ovals do overlap (do not overlap), the corresponding poles are not different (are different) at a 95% probability level. Most VGPs are in good agreement with the reference poles, which would be expected if no relative movement has occurred between the region studied and North America since the origin of the magnetization at c. 80 Ma. VGPs of sites 7, 11 and 13 diverge from this grouping. The position of VGP 7 is not well confined (see above), whereas sites 11 and 13 may reflect a local tectonic movement associated with a nearby fracture (Lugo-Hubp and Ortiz-Perez, 1980). Unfortunately, the neighbouring sites 12 and 14 (Fig. 4) gave no useful results to support this interpretation.

A mean VGP was obtained for sites 1–6 and 8–10 at 181.8° E, 76.6° N with α_{95} of 6.6°. Within the error limits, this palaeopole coincides with the 80 Ma reference pole for North America situated at 191° E, 68° N with a α_{95} of 7°. Following Beck (1980) and Demarest (1983) the apparent rotation R and apparent flattening F calculated from these data are insignificant and consequently no apparent relative movements can be deduced from the palaeomagnetic data ($R = 9° \pm 12°$, $F = 6° \pm 11°$).

The data may be interpreted in several ways. First, we are observing the effect of two tectonic movements, accidentally superimposing to bring

TABLE III

Results

Site	n	r	k	α_{95} (°)	I (°down)	D (°E)	Latitude (°N)	Longitude (°E)
1	6	1	103	6.6	46.7	334.6	65.8	186.6
2	4	1	26	18.4	50.7	333.7	64.1	194.8
3	5	1	25	15.6	31.1	347.1	77.2	149.9
4	7	0	28	11.6	33.3	5.7	85.3	39.6
5	5	0	31	14.0	−47.1	160.1	−70.3	11.9
6	5	1	41	12.1	−36.3	170.3	−80.9	344.3
7	4	1	14	25.1	43.8	287.8	24.1	184.6
8	8	0	17	14.1	45.7	356.2	82.4	228.3
9	7	0	80	6.8	41.7	342.6	73.5	180.4
10	8	0	14	15.1	36.3	344.3	75.3	166.8
11	4	1	34	15.9	−6.9	218.0	−49.3	184.1
12	5	5	n.d.					
13	5	0	23	16.3	−24.1	37.9	40.3	22.8
14	8	8	n.d					
Mean	9	3	63	6.6			76.6	181.8
Ref.				7			68	191

n, number of samples studied, r, number of samples rejected; , α_{95}, precision parameters; I, D, site mean inclination/declination; Latitude, Longitude, site mean VGP co-ordinates; Ref., reference pole for stable North America according to Irving and Irving (1982) for 80 Ma.

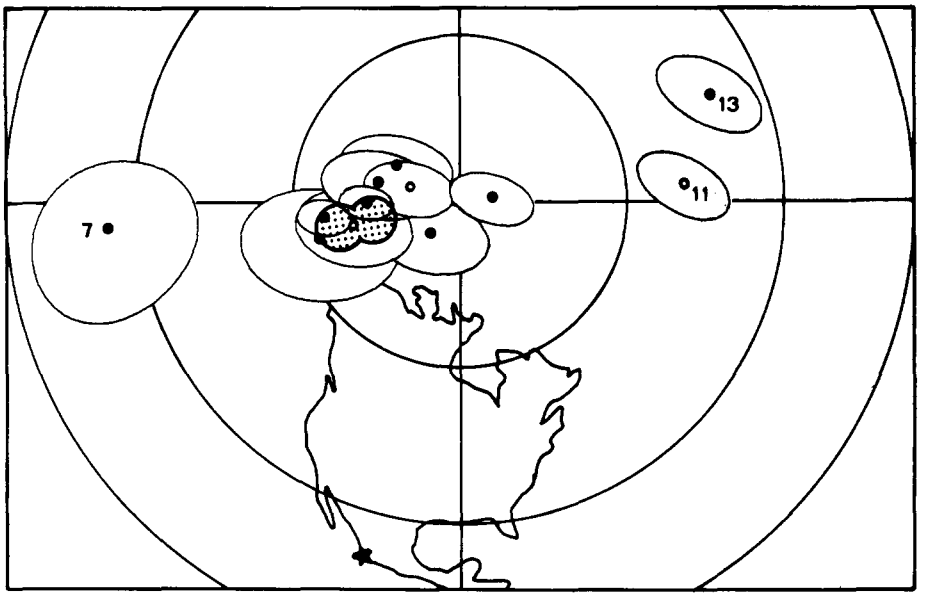


Fig. 7. Site mean VGPs and the 80 and 70 Ma reference poles for North America, with 63% error limits. ★ marks the sampling region, numbers refer to sites as given in Tables II and III.

observed and expected directions into concordance. Theoretically possible, although highly improbable, would be an accretion from any place in the Pacific and a subsequent tilting of the batholith. Second, the batholith was emplaced after accretion of the Guerrero terrain and suffered no significant movement relative to the North America Plate. This would restrict the accretion of the terrain to times before 80 Ma age and would be a serious contradiction with the suture time of Guerrero to the continent tentatively proposed by Campa and Coney (1983). Third, the batholith was emplaced in a volcanic-arc system and later accreted to the continent by purely eastward motion, which is not recognizable from palaeomagnetic data. In this case we would expect a deformation of the accreted terrain as well as of the continental margin, resulting in rotation of blocks to a different degree and with a different sense. Preliminary palaeomagnetic data from batholiths and volcanic rocks of similar age to the Puerto Vallarta intrusive complex obtained along the continental margin down to Acapulco do not point to such tectonic movements (Böhnel, 1985). We propose therefore tectonic stability of the Guerrero terrain since at least 80 Ma ago. This is also in better agreement with the geochemical data, pointing to emplacement of the batholith at an active continental margin.

Acknowledgements

This study was supported by funds of the Deutsche Forschungsgemeinschaft (Un 29/29). We thank the reviewers for their comments.

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